Ashish Shukla

BSBE Department, IIT Bombay, Mumbai, India • +91 8840603809 • ashnita16@gmail.com
https://www.linkedin.com/in/ashukla9/

PROFESSIONAL SUMMARY

Ph.D. in Biomedical Engineering from IIT Bombay specializing in photonic and optical fiber–based sensing. Experienced in nanomaterial-functionalized photonic biosensors and optical instrumentation for environmental detection.

TECHNICAL SKILLS

- Fiber-Optic Sensor Design & Photonics
- Plasmonics & Nanomaterials in Sensing
- Biofunctionalization
- Surface chemistry
- Aptamer-Based Sensor Design & Optimization
- Biosensing & Chemical Sensing
- 3D printing and Laser cutting
- MATLAB & Ansys Lumerical mode solutions
- Python
- Solidworks

WORK EXPERIENCE

Research Associate, Indian Institute of Technology Bombay **Apr 2025 - Oct 2025**

- Characterization studies for fiber-optic aptasensor.
- Nanoparticle parameter optimization studies for colorimetric arsenic detection.
- Patenting of fiber-optic aptasensor work.

Research Assistant, Indian Institute of Technology Bombay **July 2019 - July 2024**

- Managed analytical instrumentation (DLS, CD Spectropolarimeter, UV-Vis) for sample analysis and characterization.
- Conducted theory sessions and assisted in end semester examination for bachelor students.

Assistant Professor, Pranveer Singh Institute of Technology **Sept 2018 - May 2019**

- Taught undergraduate courses, delivering theoretical and practical knowledge.
- Conducted laboratory sessions, guiding students through hands-on experiments.
- Performed exam invigilation duties, ensuring academic integrity and smooth examination processes.

EDUCATION

PhD in Bio-Medical Engineering (CGPA: 8.21) **July 2019 - Oct 2025**
Indian Institute of Technology Bombay

- Thesis on "Arsenic Oxyanion Detection Scheme for Liquid Sample Matrix".
- Relevant coursework in biosensors, medical sensors, sensors in Instrumentation, etc.
- Managed analytical instrumentation (DLS, CD Spectropolarimeter, UV-Vis) for sample analysis and characterization.
- Conducted theory sessions and assisted in end semester examination for bachelor students.

M.Tech in Instrumentation Engineering (CGPA: 8.78) **July 2015 - May 2017**
National Institute of Technology Agartala

- Thesis on "Multimode Interference Based Highly Sensitive Refractive Index Sensor by Shining Optical Vortex Beam".
- Assisted in lab sessions for bachelor students.

B.Tech in Electronics and Instrumentation (Percentage: 63.82) **Aug 2010 - June 2014**
Greater Noida Institute of Technology
- Relevant coursework in Electronics and Instrumentation.

POSITION OF RESPONSIBILITIES

- **Hostel 18 Technical Secretary, IIT Bombay (Sept, 2023 - Sept, 2024)** - Led and coordinated institute-wide and hostel-level technical events, including Jarvis (securing 3rd overall position) and Howlfest, engaging over 1,300 students; additionally, established a Tech Room from scratch, enhancing IT infrastructure for the hostel.
- **Teaching Assistant for Medical Sensors and Introduction to Biology Course (IIT Bombay)** – Assisted in teaching and practical demonstrations.

PUBLICATION/PROCEEDINGS

- **Ashish Shukla**, Tathagata Pal and Soumyo Mukherji, "Polyaniline Coated U-Bent Optical Fiber Aptasensor for Detection of Arsenic in Environmental Matrices," in IEEE Sensors Journal, doi: 10.1109/JSEN.2024.3465876.
- **Ashish Shukla**, Tathagata Pal and Soumyo Mukherji, "Gold Nanoparticles Coated Aptamer-based Fiber Optic LSPR Biosensor for Arsenic Detection," 2023 IEEE Applied Sensing Conference (APSCON), Bengaluru, India, 2023, pp. 1-3, doi: 10.1109/APSCON56343.2023.10101266.
- Tathagata Pal, **Ashish Shukla**, and Soumyo Mukherji, "Development of enzyme based plasmonic optical fiber biosensor for detection of pesticide," in JSAP-Optica Joint Symposia 2022 Abstracts, (Optica Publishing Group, 2022), paper 20a_C304_9.
- Arijit Datta, Ardhendu Saha, and **Ashish Shukla**, "Investigation of modal-interference-induced fiber optic refractive index sensor: markedly enhanced sensitivity realized by shining an optical vortex beam," J. Opt. Soc. Am. A 34, 2034-2045 (2017).

INTERNATIONAL CONFERENCES

JSAP-Optica Joint Symposia, Niigata, Japan | 16th – 20th September 2024
- Oraly Presented my work titled as “Polyaniline coated U-bent Fiber Optic Aptasensor for Arsenite Detection in Environmental Matrices”.

IEEE Applied Sensing Conference, Bengaluru, India | 23rd – 25th January 2023
- Presented a Poster with title “Gold Nanoparticles Coated Aptamer-based Fiber Optic LSPR Biosensor for Arsenic Detection”. DOI: 10.1109/APSCON56343.2023.10101266

PATENT

Title: An aptasensor for detection of arsenic, and method for fabrication thereof
- India Patent No: 569256
- Grant date: 29th July 2025
- Application No: 202421040321

REFERENCES

Prof. Soumyo Mukherji
BioSensors and BioInstrumentation Lab,
Lab 301, Bioscience and Bioengineering
Department, IIT Bombay (On lien as
Director, BITS Pilani Hydrabad)
Phone: +91-22-2576 7767
Email: mukherji@iitb.ac.in

Prof. Suparna Mukherji
Treatment Remediation and Applied
MicrobiologyLab (TRAM), Lab 513,
Environmental Science dept. IIT Bombay
Phone: +91-22-25767854
Email: mitras@iitb.ac.in